Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/31/2006
PI Name:	Stein, T. Peter Ph.D.		
Project Title:	Glucose kinetics and muscle atrophy		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Physiology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	08084	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-02
Start Date:	09/01/1999	**End Date:**	08/01/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	In response to decreased usage, skeletal muscle undergoes an reductive remodeling due to the decrease in tension on the weight bearing components of the musculo-skeletal system. Accompanying this reductive remodeling of skeletal muscle in all of these situations is a decreased ability to oxidize fatty acids. The reason is not known. Possible mechanisms for the fuel shift are: (i) a shift in fuel utilization pattern involving the up regulation of glycolysis and the down regulation of fatty acid oxidation, (ii) a change at a specific locus leading to a decreased ability to oxidize fat (e.g., reduction in the carnitine acyltransferase cycle), (iii) a general decrease in metabolic activity since energy needs of inactive muscle are necessarily less and so excess fuel substrates (triglycerides, glycogen) are stored rather than used and (iv) no effect at the enzyme level. The objectives of this project are to distinguish between these three using a combination of human bed test study of glucose kinetics during bed rest and rodent hind limb unloading (HLS) studies. Work during the past year involved using the miroarray gene expression analysis with the Affymetrix gene chip system to investigate the effect of HLS on fuel		

	pathways in metabolism in rodent soleus muscle and liver. The muscle study has been completed. The principle finding was Disuse atrophy is associated with a down regulation of the total fatty acid oxidation pathway and upregulation of the glycolytic pathway.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In response to decreased usage, skeletal muscle undergoes a reductive remodeling due to the decrease in tension on the weight bearing components of the musculo-skeletal system. Accompanying this reductive remodeling of skeletal muscle in all of these situations is a shift in fuel metabolism away from lipid fuels. A frequent ancillary finding is the replacement of muscle protein with fat or glycogen. This adaptive response appears to be a general response to of muscle to disuse. It has also be found with bed rest and aging. Understanding why fuel substrate accumulation occurs could hasten recovery after bed rest and address the issue of why similar changes in muscle composition occur with aging. Underlying many pathological changes is often a metabolic change. For the elderly, it is another facet of the aging of skeletal muscle which appears to serve no benefit.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)